

Marketing Campaign Performance

Facebook Ads Dataset Analysis Report

Dataset: KAG_conversion_data.csv

Records: 1,143 ads | 3 Campaigns | 40 Interest Categories

Age Groups: 30-34 / 35-39 / 40-44 / 45-49 | Gender: M / F

Key Metrics

Metric	Value
Total Spent	\$58,705.23
Total Impressions	213,434,828
Total Clicks	38,165
Total Conversions	3,264
Approved Conversions	1,079
Overall CPC	\$1.5382
Overall CPM	\$0.28
Overall CTR	0.02%
Conversion Rate	8.55%
Cost per Conversion	\$17.99
Approval Rate	33.1%

Executive Summary

- This dataset contains 1,143 Facebook ad records across 3 campaigns.
- Total ad spend is \$58,705.23, generating 213,434,828 impressions, 38,165 clicks, and 3,264 conversions.
- The overall cost per conversion is \$17.99, with an average CTR of 0.02%.
- Campaign 1178 dominates spend (94.8% of total) but has the lowest conversion rate at 7.4%.
- Female audience generates more clicks and conversions overall, but male audience has a lower cost per conversion (\$14.94 vs \$20.99).
- Age group 30-34 has the lowest cost per conversion (\$10.66), while 45-49 has the highest (\$30.34).
- Interest category targeting shows significant efficiency variation, with cost per conversion ranging widely across categories.

Key Findings

Finding 1: Campaign Efficiency Gap

Data: Campaign 916 has a conversion rate of 51.3% and cost per conversion of \$2.58, while Campaign 1178 has only 7.4% conversion rate at \$20.86 per conversion.

Action: Action: Reallocate budget from low-efficiency campaigns to high-efficiency ones, and investigate why Campaign 1178's volume does not translate to proportional conversions.

Finding 2: Age Group Performance Variation

Data: Age group 30-34 achieves the lowest cost per conversion (\$10.66) with 1,431 conversions. Age group 45-49 has the highest spend (\$20,751) but the highest cost per conversion (\$30.34).

Action: Action: Optimize ad creative and bidding strategy for 45-49 age group, or shift budget toward 30-34 which shows better ROI.

Finding 3: Gender Differences in Efficiency

Data: Male audience has a lower cost per conversion (\$14.94) despite lower CTR (0.01% vs 0.02%).

Action: Action: Investigate whether male audience converts at higher quality (approved rate check), and whether female audience targeting can be refined to reduce wasted spend.

Finding 4: Interest Targeting Optimization Opportunity

Data: Interest categories show 2-3x variation in cost per conversion. Top-spend categories are not necessarily the most efficient.

Action: Action: Increase budget for high-conversion, low-cost-per-conversion interest categories. Reduce or pause low-efficiency categories.

Thinking Models

1. Decomposition (Breakdown)

Evidence: Total spend of \$58,705.23 was broken down by campaign, age group, gender, and interest category. This revealed that Campaign 1178 dominates spend (94.8%) but has the lowest conversion efficiency, a pattern invisible at the aggregate level.

Takeaway: Decomposition is the most fundamental analytical move: always break a total into meaningful categories before drawing conclusions.

2. Segment Divergence (Layer Differences)

Evidence: CTR ranges from 0.01% to 0.02% across age groups, and cost per conversion ranges from \$10.66 to \$30.34. Averages alone would mask these 2-3x differences.

Takeaway: Never trust the average. Always check how much segments vary around it.

3. Leverage Point Identification

Evidence: Age 30-34 Female is the highest conversion segment (619 conversions) with moderate cost per conversion (\$12.30). Improving this segment's targeting accuracy directly impacts overall campaign ROI.

Takeaway: A leverage point = high share x high improvement potential. Find where these intersect.

4. Proxy Inference

Evidence: Approved conversion rate (33.1%) serves as a proxy for conversion quality. A segment with many conversions but low approval rate may need better targeting.

Takeaway: When direct quality metrics are unavailable or slow, use approval/completion rates as a proxy.

5. Constraint vs. Preference

Evidence: Age group differences in cost per conversion may reflect product-market fit (preference) rather than campaign setup (constraint). High spend on 45-49 could be because the product appeals to them, or because campaign settings over-target them.

Takeaway: Differentiate whether the audience chooses the product or the campaign forces it. The answer determines whether to change product positioning or ad settings.

Self-Check Questions

Q1: What total metric can be decomposed, and by which dimensions?

Answer: Total spend (\$58,705) can be decomposed by campaign (3 campaigns), age group (4 groups), gender (2 groups), and interest category (40 categories).

Q2: How large are subgroup differences after decomposition?

Answer: Cost per conversion varies from \$10.66 (30-34) to \$30.34 (45-49) across age groups -- a 2.8x difference. Campaign conversion rate varies from 7.4% to 51.3%.

Q3: What missing key information can be inferred by a proxy variable?

Answer: Approved conversion rate (33.1%) is a proxy for conversion quality. Revenue per conversion is not in the dataset, so approval rate is the closest signal of whether a conversion was valuable.

Q4: Is the observed difference a preference or a constraint?

Answer: Male vs female differences in cost per conversion could be preference (product appeals more to male buyers after clicking) or constraint (ad creative is shown to more female users regardless of intent). A/B testing with controlled audience segments would help differentiate.

Q5: Which segment is both high-share and high-difference (a leverage point)?

Answer: Age 30-34 Female: 619 conversions (highest single segment) with moderate cost per conversion. Optimizing this segment's conversion rate even by 10% yields ~62 additional conversions. This is the segment with the highest leverage.

Action Recommendations

1. Rebalance Campaign Budget

Action: Shift budget from Campaign 1178 (\$55,662 spend, 7.4% conversion rate) toward Campaign 936 (27.1% conversion rate) and Campaign 916 (51.3%).

Metric to Track: Campaign-level conversion rate

Data Needed: Campaign-level spend and conversion data

2. Optimize Age 45-49 Targeting

Action: Review ad creative and landing page for 45-49 audience (\$30.34 per conversion vs \$10.66 for 30-34). Either improve relevance or reduce budget allocation.

Metric to Track: Cost per conversion for 45-49 segment

Data Needed: Age-segmented conversion and spend data

3. Scale Efficient Interest Categories

Action: Increase budget for interest categories with below-average cost per conversion (e.g., interest 15, 29). Reduce spend on categories with high cost per conversion.

Metric to Track: Interest-level cost per conversion

Data Needed: Interest-category spend and conversion reports

4. A/B Test Gender Creative

Action: Test separate ad creatives for male (\$14.94/conv) and female (\$20.99/conv) audiences to determine whether the cost difference is driven by creative or audience fit.

Metric to Track: Gender-segmented CPC and conversion rate

Data Needed: A/B test results with controlled creative variables

Metrics and Concepts Explained

CPC (Cost Per Click)

Meaning: Cost per individual click on an ad.

Formula: $\text{Spent} / \text{Clicks} = 58705.23 / 38165 = \1.5382

Business Use: Measures how efficiently budget generates clicks. Lower is better for traffic-oriented campaigns.

CPM (Cost Per Mille)

Meaning: Cost per 1,000 impressions.

Formula: $(\text{Spent} / \text{Impressions}) \times 1000 = (58705.23 / 213434828) \times 1000 = \0.28

Business Use: Used for brand awareness campaigns where visibility matters more than clicks.

CTR (Click-Through Rate)

Meaning: Percentage of impressions that resulted in a click.

Formula: $(\text{Clicks} / \text{Impressions}) \times 100 = (38165 / 213434828) \times 100 = 0.02\%$

Business Use: Measures ad relevance. Higher CTR generally means the creative matches the audience.

Conversion Rate

Meaning: Percentage of clicks that resulted in a conversion.

Formula: $(\text{Total_Conversion} / \text{Clicks}) \times 100 = (3264 / 38165) \times 100 = 8.55\%$

Business Use: Measures post-click effectiveness. Low conversion rate suggests landing page or offer issues.

Cost per Conversion (CPA)

Meaning: Average cost to acquire one conversion.

Formula: $\text{Spent} / \text{Total_Conversion} = 58705.23 / 3264 = \17.99

Business Use: The most important ROI metric. All optimization efforts should ultimately reduce this.